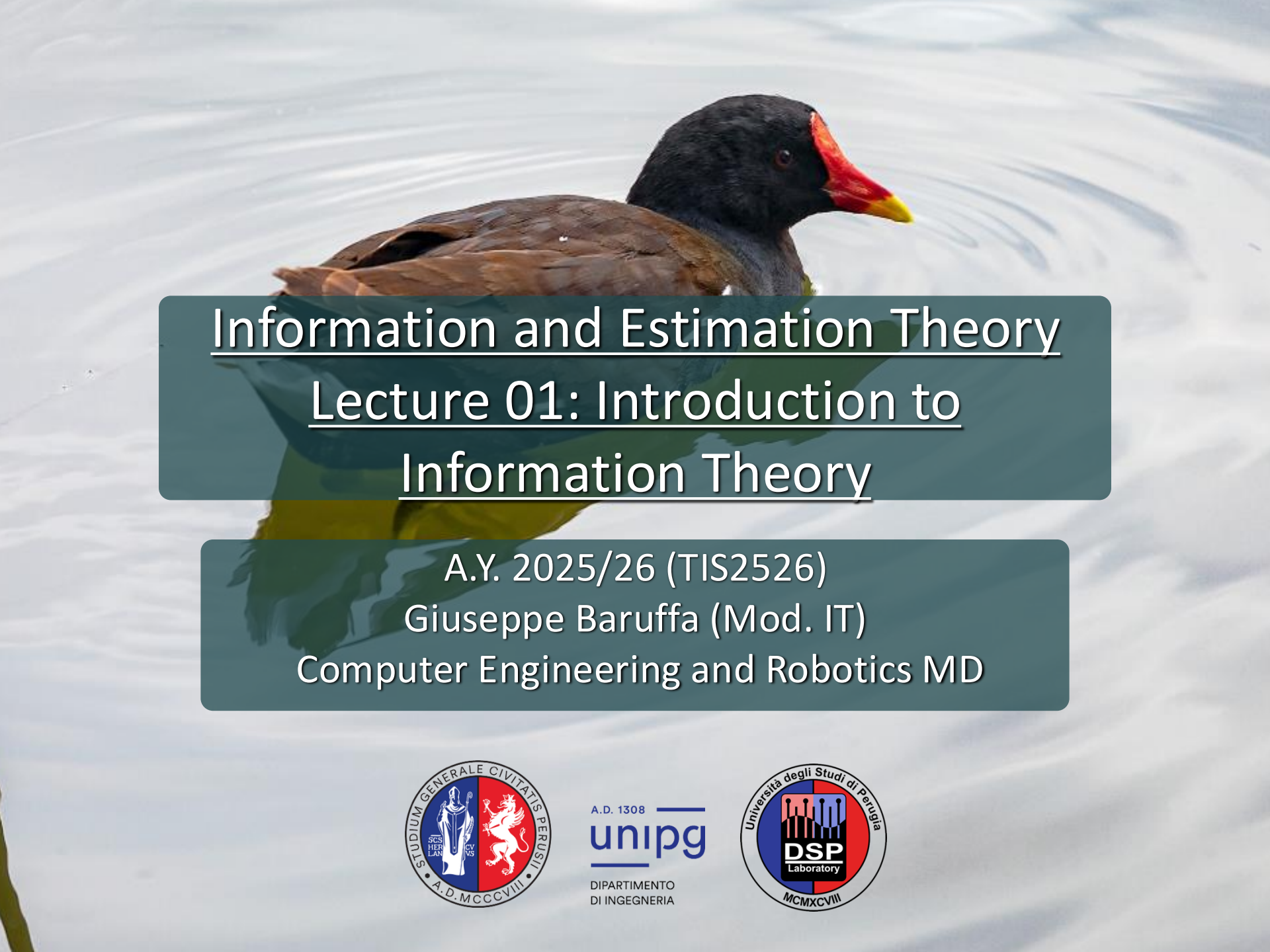


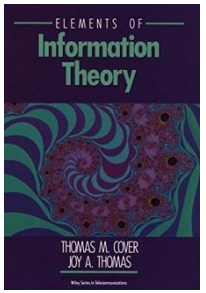
Information and Estimation Theory
Lecture 01: Introduction to Information Theory

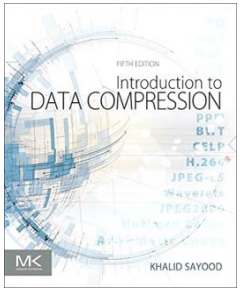
A.Y. 2025/26 (TIS2526)
Giuseppe Baruffa (Mod. IT)
Computer Engineering and Robotics MD

STUDIUM GENERALE CIVITATIS PERUSII A.D. MCCCVIII
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Università degli Studi di Perugia DSP Laboratory MCMXCVIII

The module in a nutshell

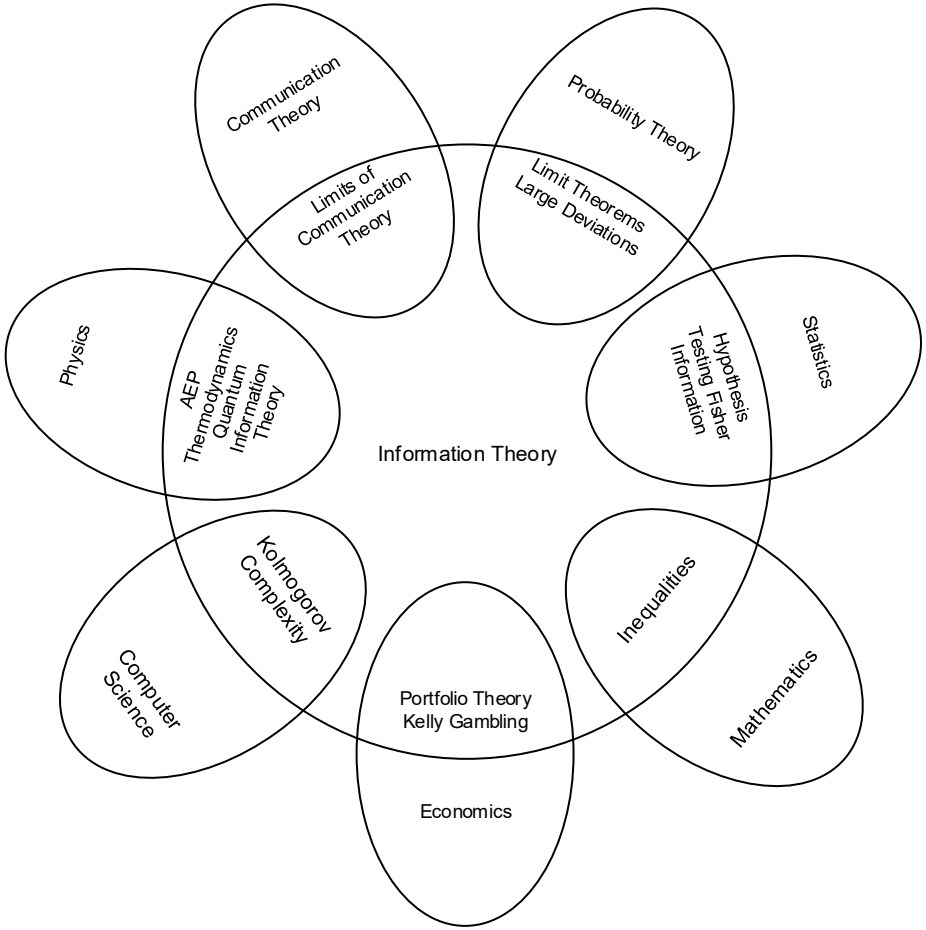
- Syllabus for Module «Information Theory» (IT) of TIS2526
  - Introduction to information theory
    - Measure of information. Entropy, relative entropy, and mutual information. Relation between entropy and mutual information. Chain rule for entropy, relative entropy, and mutual information. Jensen inequalities, data processing and log-sum. Fano's inequality. Asymptotic equipartition property. Entropy rates for discrete stochastic processes. Differential entropy.
  - Source codes
    - Classes of source codes. Kraft inequality and its consequences. Optimal codes: Huffman codes. Shannon-Fano-Elias codes. Universal codes: arithmetic coding, Lempel-Ziv coding. Elements of rate-distortion theory (RD): definition, RD functions.
  - Reference bibliography
    - Thomas, M. Cover and Joy A. Thomas. Elements of information theory (2nd ed.). Wiley-Interscience, 2006
    - Khalid Sayood. Introduction to data compression (5th ed.). Morgan-Kaufmann, 2017





An overview

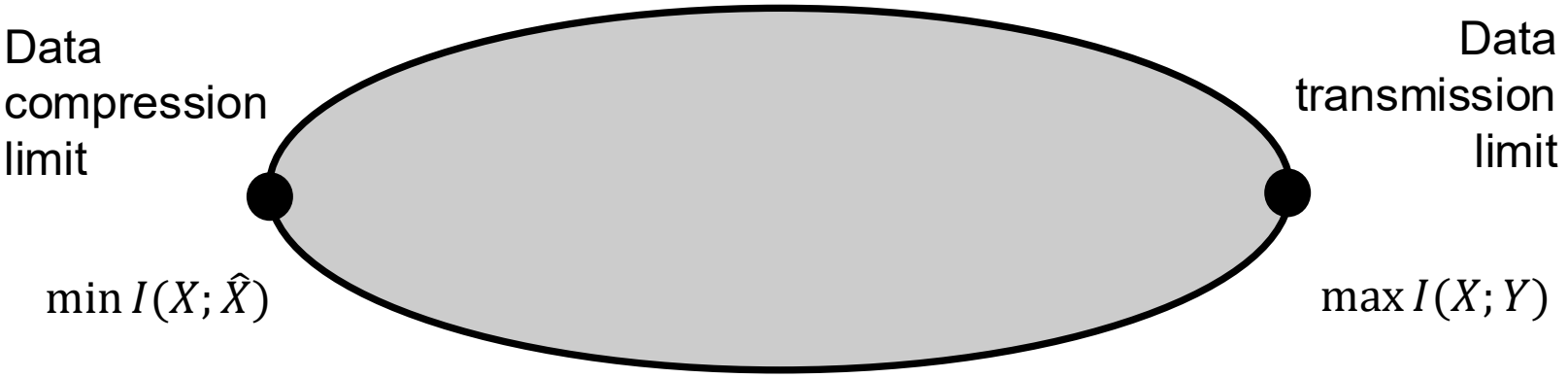
- Information theory answers two fundamental questions
  - What is the ultimate data compression? (**entropy *H***)
  - What is the ultimate rate of communication? (**capacity *C***)
- Fundamental contributions to
  - statistical physics (*thermodynamics*), computer science (*Kolmogorov*), statistical inference (*Occam's Razor*), probability and statistics (*hypothesis testing*)



- **Intersects**: physics (statistical mechanics), mathematics (probability theory), electrical engineering (communication theory), computer science (algorithmic complexity)

Electrical Engineering (Communication Theory)

- In the ‘40s, Shannon proved that the probability of error could be almost nulled for communication rates below the channel **capacity $C = I(X;Y)$**, which depends on the channel noise
- Shannon argued that random processes (e.g., audio) have a complexity limit below which the signal cannot be compressed: the **entropy $H = I(X;\hat{X})$**, with reference to thermodynamics
- If the entropy is less than the capacity, asymptotically error-free communication can be achieved



Computer Science (Kolmogorov Complexity)

- Kolmogorov, Chaitin, and Solomonoff proposed that the complexity *K* of a data string is the length of the **shortest binary computer program** for computing the array
  - This definition of complexity (minimal description length) is universal (e.g., computer independent) and fundamental
- This complexity *K* approximates the entropy *H* if the sequence is drawn at random from a distribution that has entropy *H*
- Kolmogorov complexity ***K* is more fundamental than entropy *H***
  - Ultimate data compression and logically consistent inference procedure
- Computational complexity ***O* (time complexity)** and Kolmogorov complexity ***K* (descriptive complexity)** are two axes corresponding to program running time and length
  - Kolmogorov complexity focuses on minimizing along the 2nd axis, and computational complexity focuses on minimizing along the 1st axis

Physics and Mathematics

- *Thermodynamics*
  - Statistical mechanics is the birthplace of entropy and the second law of thermodynamics
  - Entropy always increases
  - The second law dismisses any claims to perpetual motion machines
- *Probability Theory and Statistics*
  - The fundamental quantities of information theory (entropy, relative entropy, and mutual information) are defined as functionals of probability distributions
  - In turn, they characterize the behavior of long sequences of random variables and allow us to estimate the probabilities of rare events (large deviation theory) and to find the best error exponent in hypothesis tests

Philosophy of Science (Occam's Razor)

- William of Occam said "*Numquam ponenda est pluralitas sine necessitate*", i.e., "The simplest explanation is best"
- Solomonoff and Chaitin argued that a universally good prediction procedure takes a weighted combination of all programs for the data and observes what prints next
  - E.g., this procedure will eventually predict the subsequent digits of π, or coin flips that come up heads with probability 0.7
- Of course, such inference is highly impractical, because discarding all computer programs that fail to generate existing data will take impossibly long
  - It would take a hundred years to predict what happens tomorrow

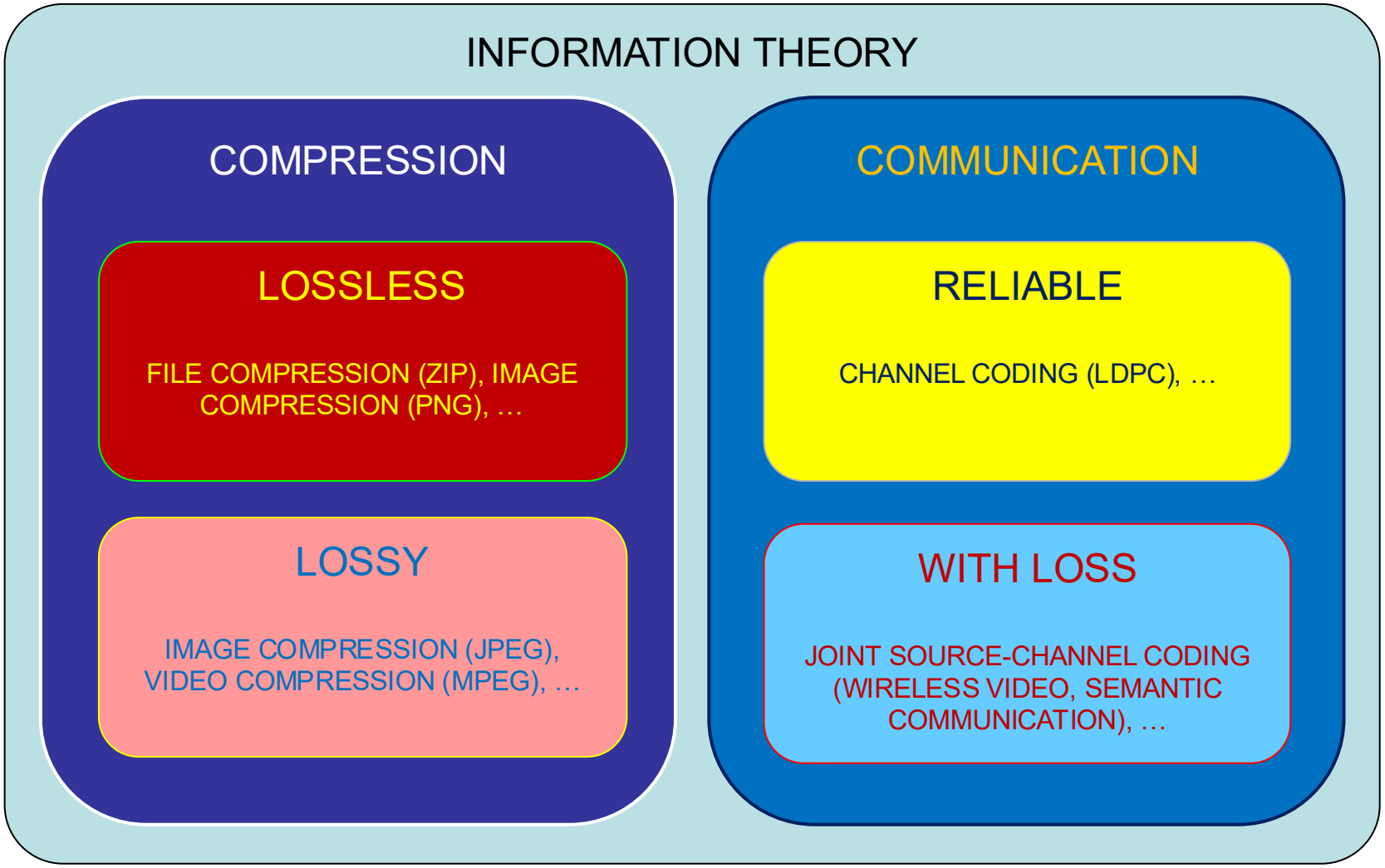


Economics and Computation

- Economics (Investment)
  - Repeated investment in a stationary stock market results in an exponential growth of wealth
  - The wealth growth rate is a dual of the *entropy rate* of the stock market
  - There are parallels between the theory of optimal investment in the stock market and information theory
- Computation vs. Communication
  - As we build larger computers out of smaller components, we encounter both a computation limit and a communication limit
  - Computation is communication-limited and communication is computation-limited
  - These become intertwined, and thus all the developments in communication theory via information theory should have a direct impact on the theory of computation

Topics of Information Theory

- Succinctly



Additional credits

Slides, examples, images, books, papers, …

- EE376A/STATS376A Information Theory – T. Weissman (Stanford)
- Shao, Yulin, Qi Cao, and Deniz Gunduz. "A theory of semantic communication." arXiv preprint arXiv:2212.01485 (2022).
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- https://commons.wikimedia.org/wiki/File:Abejaruco_sobre_higuera.jpg
- https://xkcd.com/2582/
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- https://commons.wikimedia.org/wiki/File:2015-10-19_Botaurus_stellaris,_Gosforth_Park_2.jpg
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- https://commons.wikimedia.org/wiki/File:Olivetti_Lettera_22_by_LjL.jpeg
- https://commons.wikimedia.org/wiki/File:Aegithalos_caudatus_side-on.jpg
- https://commons.wikimedia.org/wiki/File:Morse_code_in_British_signal_exchange_Somme_1916_IWM_Q_128.jpg
- https://commons.wikimedia.org/wiki/File:Lettres_en_code_Morse_fr_en.png
- https://commons.wikimedia.org/wiki/File:Eurasian_Wryneck_(Jynx_torquilla)_(32837517738).jpg
- https://commons.wikimedia.org/wiki/File:20_questions_1954.JPG
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- K. Sayood, *Introduction to data compression*, 3rd ed., Morgan Kauffman, 2006
- Tishby, Naftali, Fernando C. Pereira, and William Bialek. "The information bottleneck method." arXiv preprint physics/0004057 (2000)
- Tishby, Naftali, and Noga Zaslavsky. "Deep learning and the information bottleneck principle." 2015 ieee information theory workshop (itw). IEEE, 2015
- https://upload.wikimedia.org/wikipedia/commons/5/59/Tetrao_urogallus%2C_Glenfeshie%2C_Scotland_1.jpg
- JPEG Committee, “JPEG AI the new image compression standard entirely based on neural networks”, Dr. ELENA ALSHINA, Huawei Technologies, Germany
- Ascenso, Joao, Elena Alshina, and Touradj Ebrahimi. "The JPEG AI standard: Providing efficient human and machine visual data consumption." *IEEE Multimedia* 30.1 (2023): 100-111.
- Kevin M. Murphy, Probabilistic Machine Learning: An introduction, MIT Press, 2022